Fiscal Year:	FY 2019	**Task Last Updated:**	FY 11/29/2019
PI Name:	Beeson, Harold Ph.D.		
Project Title:	Fundamental Research on International Standard of Fire Safety in Space - Subteam 5: Evaluation of Test Methods for Practical Materials		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	harold.d.beeson@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	550-524-5542
Organization Name:	NASA Johnson Space Center (JSC) White Sands Test Facility (WSTF)		
PI Address 1:	12600 NASA Road		
PI Address 2:	Bldg. 120		
PI Web Page:			
City:	Las Cruces	**State:**	NM
Zip Code:	88012	**Congressional District:**	2
Comments:	NOTE: Retired as of May 2019 (per S. Olson/GRC)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	07/01/2014	**End Date:**	03/31/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
Contact Email:	david.l.urban@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: PI Beeson retired in March 2019; project transferred to Jonathan Tylka as PI, per S. Olson/GRC (Ed., 4/4/19)		
Key Personnel Changes/Previous PI:	Harold Beeson is U.S. Co-Investigator on Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA Principal Investigator (PI) is Prof. Osamu Fujita, Hokkaido University.		
COI Name (Institution):	Hirsch, David Ph.D. (NASA Johnson Space Center/White Sands Test Facility) Harper, Susana B.S. (NASA White Sands Test Facility (WSTF)) Juarez, Alfredo M.S. (Jacobs @ NASA White Sands Test Facility (WSTF)) Woods, Brenton B.S. (ERC @ NASA White Sands Test Facility (WSTF))		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	ED. NOTE (April 2019): Principal Investigator Harold Beeson retired in March 2019; project transferred to Jonathan Tylka as PI, per S. Olson/NASA Glenn Research Center. See that project ("Fundamental Research on International Standard of Fire Safety in Space - Subteam 5: Evaluation of Test Methods for Practical Materials--PI Tylka") for future reporting. Funding is for Dr. Beeson's role as U.S. Co-Investigator for the Japan Aerospace Exploration Agency (JAXA)-sponsored project, “Flammability Limits At Reduced-g Experiment (FLARE)." JAXA International Announcement of Opportunity (AO) to fund experiments to be conducted aboard the Japanese Experiment Module, Kibo, 2012. NASA’s fire safety standards have not confirmed scientifically that they are always conservative for fire safety in space because the test is based on the flammability in the normal gravity. It is already known that a solid material’s flammability in microgravity may be greater than in normal gravity for some conditions. Therefore, whether the NASA's standards will be applied for future space missions by individual countries, or if better general standards for fire safety (e.g., oxygen index method) would be better for fire safety in space should be studied. This work aims to determine what standards are adequate for the evaluation of fire safety and material flammability in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Studying materials flammability in space allows us to accurately control the flow field and thus elucidate the importance of a critical Damkohler number (flow time /reaction time) on flame extinction. The anticipated improved methodology should reduce time and cost for the spacecraft material screening. Investigation and results have Earth benefits for terrestrial fire safety.
Task Progress:	ED. NOTE (April 2019): Principal Investigator (PI) Harold Beeson retired in March 2019; project transferred to Jonathan Tylka as PI, per S. Olson/NASA Glenn Research Center. See that project ("Fundamental Research on International Standard of Fire Safety in Space - Subteam 5: Evaluation of Test Methods for Practical Materials--PI Tylka") for future reporting.
Bibliography Type:	Description: (Last Updated:)